DO NOW · AO1 REVISION

Pressure in fluids and in the atmosphere

Forces
Paper 2 · Separate Physics only
17 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the two states of matter that are considered fluids.**	Liquid and gas	☐
2	**Describe what causes pressure in a fluid.**	The particles of the fluid colliding with surfaces.	☐
3 ★	**A fluid exerts a force on its container. State the direction of the force.**	At right angles to the surface.	☐
4	**State the equation linking pressure, area and force.**	pressure = force ÷ area (p = F ÷ A)	☐
5 ★	**State the standard unit of pressure.**	Pascal (Pa)	☐
6 ★	**Describe how the pressure in a liquid depends on the height of the liquid column.**	The taller the column, the greater the pressure.	☐
7	**Describe how the pressure in a liquid depends on its density.**	The denser the liquid, the greater the pressure.	☐
8 ★	**Explain what causes upthrust.**	Pressure increases with depth, so the pressure on the bottom of a submerged object is greater than on the top.	☐
9	**Describe the Earth's atmosphere.**	A thin layer of air around the Earth.	☐
10	**Describe how the density of the atmosphere changes with height.**	It decreases as height increases.	☐
11	**Give the cause of atmospheric pressure.**	Air molecules colliding with a surface.	☐
12	**Describe how atmospheric pressure changes with height.**	It decreases as height increases.	☐

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	QUESTION	ANSWER	✓
13	**State the equation for the pressure due to a column of liquid.**	pressure = height × density × gravitational field strength ($p = h \rho g$)	☐
14 ★	**Explain why the pressure in a liquid increases with depth.**	The deeper you go, the greater the weight of liquid above.	☐
15	**State the condition for an object to float.**	Upthrust equals its weight.	☐
16	**State when an object in a liquid sinks.**	When its weight is greater than the upthrust.	☐
17 ★	**Explain why atmospheric pressure decreases as height increases.**	There is less air above, so fewer air molecules collide with each unit of area.	☐

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